Fields

$\mathbb{Q}$ set of rational numbers $\qquad \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

$\mathbb{R}$ set of real numbers

$\mathbb{C}$ set of complex numbers

common

$+$ addition $\qquad 0$ additive identity

$\cdot$ multiplication $\qquad 1$ multiplicative identity

If $a \in \mathbb{Q}$ $\exists$ $-a$ s.t. $a+(-a)=0$ (additive inverse)

If $0 \neq a \in \mathbb{Q}$ $\exists$ a^{-1} st $a \cdot a^{-1} = 1$ (multiplicative inverse)

non-field $\mathbb{Z}$ $\qquad 2 \in \mathbb{Z}$ there is no multiplicative inverse of 2 in $\mathbb{Z}$.

$M_n(\mathbb{R})$ $n \times n$ real matrizes

Def A field F is a set with two binary operations $+$ & $\cdot$ satisfying:

1) $(F, +)$ is an abelian group

☆ 2) $(F - \{0\}, \cdot)$ is ~~a~~ an abelian group

3) Distributive Law $a \cdot (b+c) = a \cdot b + a \cdot c$

Eg) $\mathbb{Q}$ $a + (-a) = 0$

$a \cdot \frac{1}{a} = 1$

Eg) $\mathcal{F} = \left\{ \frac{g(x)}{h(x)} : g(x), h(x) \in \mathbb{R}[x] \right\}$

polynomials with coefficients in $\mathbb{R}$.

$\bullet$: $\frac{g_1(x)}{h_1(x)} \bullet \frac{g_2(x)}{h_2(x)} = \frac{g_1(x) \cdot g_2(x)}{h_1(x) \cdot h_2(x)} \in \mathcal{F}$

$+$: $\frac{g_1(x)}{h_1(x)} + \frac{g_2(x)}{h_2(x)} = \frac{g_1(x)\, h_2(x) + g_2(x)\, h_1(x)}{h_1(x) \cdot h_2(x)} \in \mathcal{F}$

$0 = \frac{0}{1}$ addative identity

$\left(\frac{g(x)}{h(x)}\right)^{-1} = \frac{h(x)}{g(x)}$ mult inverse

$-\left(\frac{g(x)}{h(x)}\right) = \frac{-g(x)}{h(x)}$ addative inverse

$1 = \frac{1}{1}$ mult. identity

Def a <u>subfield</u> of a field F is a subset of F that forms a field under the same operations of F. If E is a subfield of F then $E \leq F$.

~~note~~ $\mathbb{Z} \subseteq \mathbb{Q}$ but $\mathbb{Z} \not\leq \mathbb{Q}$

Eg1 $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ $\qquad$ $\mathbb{Q} < \mathbb{R} < \mathbb{C}$

$\mathbb{Q}(x)$ set of rational functions w/ coeffs in $\mathbb{Q}$

$\mathbb{R}(x)$ " " " " " " " $\mathbb{R}$

$\mathbb{C}(x)$ set of rational functions w/ coeffs in $\mathbb{C}$

$\mathbb{Q} < \mathbb{Q}(x)$ $\qquad$ $\mathbb{Q}(x) < \mathbb{R}(x) < \mathbb{C}(x)$

Subfield Test

Prop 16.5 Let $E \leq F$ a field, then $E \leq F$ if and only if the following conditions hold:

1) $(E, +) < (F, +)$

2) $(E - \{0\}, \cdot) < (F - \{0\}, \cdot)$

Proof is on HW.

Example of subfield test

Let $F = \{ r(x) \in \mathbb{R}(x) : r(x) = r(-x) \}$

even rational functions

want to show $F \leq \mathbb{R}(x)$

additive structure

NTS $(F, +) < (\mathbb{R}(x), +)$

identity

$\frac{0}{1} \in F$

closure $r_1(x), r_2(x) \in F$

$r_1(-x) + r_2(-x) = r_1(x) + r_2(x)$ $\quad r_1(x) + r_2(x) \in F$

closed under inverses $r(x) \in F$

$-r(-x) = -(r(-x)) = -r(x)$ $\quad$ so $-r(x) \in F$

multiplicative structure

mult. identity $\quad \frac{1}{1} \in F - \{0\}$

closed under mult $\quad r_1(x), r_2(x) \in F - \{0\}$

$r_1(-x) \cdot r_2(-x) = r_1(x) \cdot r_2(x) \implies r_1(x) \cdot r_2(x) \in F - \{0\}$

closed under inverse $\quad r(x) \in F$

$$\frac{1}{r(-x)} = \frac{1}{r(x)} = (r(x))^{-1}$$

so $(r(x))^{-1} \in F - \{0\}$

so $(F - \{0\}, \cdot) < (\mathbb{R}(x) - \{0\}, \cdot)$

so $F \leq \mathbb{R}(x)$

Some finite Field

$\mathbb{Z}_2 = \{0, 1\}$ from before $(\mathbb{Z}_2, +)$ is an abelian group

$\mathbb{Z}_2 - \{0\} = \{1\}$ $1 \cdot 1 = 1$ trivial group

$a(b+c) = ab + ac$ $\forall$ $a, b, c \in \mathbb{Z}_2$

$a=b=c=0$ $\rightarrow 0(0+0) = 0 = 0 \cdot 0 + 0 \cdot 0$

$a=1$, $b=c=0$ $1 \cdot (0+0) = 0 = 1 \cdot 0 + 1 \cdot 0$

$1 \cdot (1+1) = 0 = 1 \cdot 1 + 1 \cdot 1$

5 more

$\mathbb{Z}_3 = \{0, 1, 2\}$

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

is $\mathbb{Z}_2$

In general, claim: $\mathbb{Z}_p$ is a field of p elements for p prime under multiplication & addition modulo p.

<u>Claim</u> $(\mathbb{Z}_n, +, \cdot)$ is a field if and only if n is prime.

$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ not a finite field

$\mathbb{Z}_6 - \{0\} = \{1, 2, 3, 4, 5\}$ abelian group under mult?

$2 \cdot 3 = 0 \bmod 6$

$2 \cdot 3 = 0$ multiplication is not closed

$\mathbb{F}_4 = \{0, 1, \alpha, 1+\alpha\}$ where $\alpha^2 = 1+\alpha$, $1+1=0$

not $\mathbb{Z}_4$

$+$	0	1	α	$1+\alpha$
0	0	1	α	$1+\alpha$
1	1	0	$1+\alpha$	α
α	α	$1+\alpha$	0	1
$1+\alpha$	$1+\alpha$	α	1	0

$\cdot$	0	1	α	$1+\alpha$
0	0	0	0	0
1	0	1	α	$1+\alpha$
α	0	α	$1+\alpha$	1
$1+\alpha$	0	$1+\alpha$	1	α

cyclic of order 3